

Safety Analysis Summary Report

AURIX 2G Family

About this document

This Safety Analysis Summary Report describes the performed safety analysis for the AURIX 2G devices and provides reference to the corresponding safety analysis results.

Scope and purpose

This Safety Analysis Summary Report is applicable for the following products:

Name	Package
TC38x	Bare Die ¹⁾ , LFBGA-292, LFBGA-516
TC39x	LFBGA-516, LFBGA-292, LFBGA-292 with special ADAS pinning
TC35x	LFBGA-180, LFBGA-292 with special ADAS pinning
TC37xED	Bare Die ¹⁾ , LFBGA-292, LQFP-176, LQFP-144
TC37xPD	Bare Die ¹⁾ , LFBGA-292, LQFP-176
TC36x	Bare Die ¹⁾ , LFBGA-292, LFBGA-180, LQFP-176, TQFP-144, LQFP-144
TC33xED	LFBGA-292 ADAS, LFBGA-180 ADAS
TC33xPD, TC32x	LFBGA-292, LFBGA-180, TQFP-144, TQFP-100, TQFP-80
TC3Ex	LFBGA-292

Note:

1. The integration and qualification of bare die microcontrollers into its application environment (including related Dependent Failures) is the responsibility of the integrator (e.g. housing, passivation, soldering temperature) to ensure ISO 26262 and IEC 61508 compliance for his target safety application.

Additional device specific detailed information and technical properties can be derived from Safety Manual [1-2] and from device specific User Manuals.

Intended audience

This document is provided under NDA to AURIX™ 2G external customers.

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1 Introduction

This Safety Analysis Summary Report describes the performed safety analysis for the AURIX 2G Family and provides reference to the corresponding safety analysis results.

This Chapter 1 provides the introduction to the Safety Analysis Summary Report.

Relevant information for the system integrator when using the safety results are described in Chapter 2

The subject of the analysis is summarized in Chapter 3

The applied safety analysis methods for the AURIX family of devices are described in Chapter 4

The Safety analysis results are summarized in following chapters

- Fault Tree Analysis (FTA), see Chapters 5.1 and 5.3
- Dependent Failure Analysis (DFA), see Chapter 5.4
- Failure Mode and Effects Analysis (FMEA), see Chapters 5.2 and 5.5
- Failure Modes Effect and Diagnostics Analysis (FMEDA), see Chapter 5.6

The safety analysis conclusion is presented in Chapter 6.

Reference documents are listed in Chapter 7 and a list of abbreviations provided in Chapter 8.

2 Scope for the System Integrator

Relevant information for the system integrator concerning the use of the safety analysis results at system level is available as listed below

- Infineon provides fully configurable FMEDAs ([1-4] to [1-12]) containing all relevant data and information. It is the responsibility of the integrator to customize the FMEDAs for a specific application use case in order to assess Failure Mode Effects and to derive SPFM, LFM, PMHF, SFF, PFH, and PFD metrics. Customization include, but not limited to, reviewing and adjusting operating profile, configuring the FMEDA for application use case, reviewing and adjusting the safety mechanisms. After customization, it is the responsibility of the integrator to review the final FMEDA results. Infineon could provide some initial support via Field Application Engineers (FAE) support structure in order for the integrator to get familiar with the FMEDA and to answer questions related to FMEDA.
- Pin and package related failure modes are also covered by FMEDAs and can be configured accordingly. In addition, package and pin-out related recommendations given in relevant Application Notes [1-13] shall be considered.
- Applications using bare die products* shall consider the alpha particle activity of the material used for encapsulation and interconnect and if necessary, scale the soft error failure rate (SER) given by AURIX FMEDA accordingly. Rationale: AURIX FMEDA considers the alpha activity caused by the mold compound of AURIX packages which is specified to be less than 10 counts per hour per square meter (10 cts/ (hour*m²)).
- The DFA analysis outcomes, including the analysis done on External Safety Mechanisms (ESM), are incorporated in the Safety Manual [1-2]. Typical and significant dependent failures are also quantified in the FMEDAs ([1-4] to [1-12]). Additional package related recommendations are documented in listed application note [1-13] and must be taken into account by the customer.
- Package and pin-out related recommendations (e.g. allocation of redundant I/O channels for avoidance of common causes) are referenced or described in application note [1-13], which shall be considered by the customer.

*Attention: * Integration and qualification of bare die microcontrollers into its application environment (including related Dependent Failures) is the responsibility of the integrator (e.g. housing, passivation, soldering temperature), to ensure ISO 26262 and IEC 61508 compliance for the target safety application.*

3 Subject of the Analysis

This section provides reference to documents providing an overview of product under assessment (AURIX 2G Family of devices).

3.1 Overview of the Product

AURIX User's manuals [1-14] provide the detailed description of Infineon AURIX TC3xx devices.

3.2 Known Bugs/ Product Limitations of AURIX 2G Family

For product limitations and known bugs, refer to the latest available Errata Sheets.

4 Safety Analysis Methods

This section summarizes the different analysis performed.

4.1 Required Safety Analysis Methods acc. to ISO26262

As per section ISO26262-4, clause 7.4.3.1, the methods given in Table 1 are applied to identify the causes of systematic failures and the effects of systematic faults during the concept development phase.

Table 1 Required safety analysis methods acc. to ISO26262-4

Methods	ASIL A	ASIL B	ASIL C	ASIL D	Method applied in AURIX 2G
1 Deductive analysis	o	+	++	++	Concept Fault Tree Analysis (CFTA)
2 Inductive analysis	++	++	++	++	Concept Failure Mode Effect Analysis (CFMEA)

As per section ISO26262-5, clause 7.4.3.1, the methods given in Table 2 are identified for safety analysis of Hardware design.

Table 2 Required safety analysis methods acc. to ISO26262-5

Methods	ASIL A	ASIL B	ASIL C	ASIL D	Method applied in AURIX 2G
1 Deductive analysis	o	+	++	++	Fault Tree Analysis (FTA)
2 Inductive analysis	++	++	++	++	Failure Mode and Effects Analysis (FMEA) Failure Mode Effects and Diagnostic Analysis (FMEDA)

Table 3 explains all safety analysis methods applied.

#	Safety Analysis Methods		Objective of the analysis
1	-	Firmware Software Safety Analysis, Hazard and Operability Analysis (SWSA / FW-HAZOP)	To address systematic faults in Firmware.
2	-	Dependent Failure Analysis (DFA-SW) on Software (FW)	To identify the single events/causes that could bypass or invalidate the required independence or freedom from interference between given elements leading to a violation of safety requirement or a safety goal. DFA-SW was performed for Firmware.
3	-	CPU SBST Software Safety Analysis, Hazard and Operability Analysis (CPU SBST HAZOP)	To address systematic faults in CPU SBST.
4	-	SPU SBST Software Safety Analysis, Hazard and Operability Analysis (CPU SBST HAZOP)	To address systematic faults in SPU SBST.
5	Inductive	Design FMEA (DFMEA)	To identify and mitigate systematic faults.

6	Inductive	Concept Failure Mode Effect Analysis (CFMEA)	This WP identifies and analyses FB Failure Modes for avoidance of random faults at a Concept Level (bottom-up approach). The analysis ensures a plausible description of each failure mode and the associated TSR to handle the respective failure mode based on the chosen monitoring method. To evaluate Start up Software for additional derived failure modes originated from Random Hardware errors occurring during boot process, which could affect any Safety functionality within the chip (FW-FMEA).
7	Inductive	Failure Mode and Effects Analysis (FMEA)	This WP (also known as <i>p5FMEA</i>) serves two purposes: 1.) Analysis of the development process for all FBs and DPs (scope of the analysis covers all DPs for the avoidance of systematic faults.) 2.) Analysis of all DPs against random hardware faults including detailed allocation of Safety Mechanisms for each failure mode at the respective DP level.
8	-	DFA ²	To identify and analyse dependent failure initiators (DFIs). To check common cause and cascading failures. DFA aims primarily to identifying the single event that could bypass or invalidate a required independence or freedom from interference between given elements.
9	Inductive	FMEDA ³	To verify a hardware design against defined targets for the evaluation of the hardware architectural metrics and the evaluation of safety goal violations due to random hardware failures To support customer for system level analysis.
10	Deductive	Concept FTA (CFTA)	To identify and analyse FB Failure Modes for avoidance of Systematic Faults at a Concept Level (Top down). To be done on SoC level down to the appropriate level of detail, where the safety monitoring functions are effective.
11	Deductive	FTA	To identify and analyse DP Failure Modes for avoidance of random faults at an Architectural Element Level (bottom-up approach). The FTA is a deductive analysis starting with the undesired system behaviour and determining the possible causes of this behaviour. The content of the FTA provides inputs for the DFA.

Table 3 Safety Analysis Methods

Note:

1. The different types of dependent failures are assessed qualitatively inside the DFA according to ISO26262-9, chapter 7. The DFA ESMs (External SW Safety Mechanisms) aims to analyze shared resources and Dependent Failure Initiators for all redundancy based safety application use cases as per the Safety Concept
2. ISO26262-5, clause 7.4.3 requires an inductive analysis method for products supporting safety applications. For the AURIX the selected inductive method is the quantitative Failure Mode Effect and Analysis (FMEA), otherwise called Failure Mode Effect and Diagnostic Analysis (FMEDA).

The performed FMEDA aims to evaluate AURIX failure rate contributions and its' according failure metrics. The objectives of the FMEDA are

- Verification of Hardware design in accordance with ISO26262-5, clause 7.4.4, Table 3
- Evaluation of single point fault metrics (SPFM) in accordance with ISO26262-5, clause 8.4.5, Table 4
- Evaluation of latent fault metrics (LFM) in accordance with ISO26262-5, clause 8.4.6, Table 5
- Evaluation of Probabilistic Metric for random Hardware Failures (PMHF) in accordance with ISO26262-5, clause 9.4.2

4.2 IEC 61508 Safety Analysis

Safety Analysis is required in order to fulfill IEC 61508 compliance, independent of SIL (Safety Integrity Level). The following Safety Analysis Methods have been performed with regards to IEC 61508. See also AURIX™ TC3xx Safety Manual IEC61508 Addendum [1-3].

Table 4 Safety Analysis Methods applied for IEC 61508

#	Safety Analysis Methods		Objective of the analysis, Comments
1	Inductive	FMEDA - Failure Mode Effect and Diagnostic Analysis	To verify a hardware design against defined targets for the evaluation of the hardware architectural metrics and the evaluation of safety goal violations due to random hardware failures.
2	Deductive	FTA – Fault Tree Analysis	To identify and analyse DP Failure Modes for avoidance of random faults at an Architectural Element Level (bottom-up approach). The FTA is a deductive analysis starting with the undesired system behaviour and determining the possible causes of this behaviour. The content of the FTA provides inputs for the DFA.
3	-	DFA – Dependent Failure Analysis	To identify and analyse dependent failure initiators (DFIs). To check common cause and cascading failures. DFA aims primarily to identifying the single event that could bypass or invalidate a required independence or freedom from interference between given elements.
4	Inductive	FMEA - Failure Mode and Effects Analysis	To identify systematic faults in the product.

5 Safety Analyses

The reference to the main safety analysis results derived from different work packages as listed in Chapter 4 are presented in following chapters. This report focusses on safety relevant information that is not available in other customer documents. Safety related recommendations derived from any Safety Analysis result are included or referenced in Safety Manual [1-2] if not stated otherwise.

5.1 CFTA

The Concept Level FTA was performed to support evaluation of Random Hardware Faults and Systematic Faults as verification measure of TSR completeness in the A2G Safety Concept.

5.2 CFMEA

A Failure Mode and Effects Analysis (FMEA) was performed on conceptual level including analysis of Startup Software (Firmware FW) of AURIX 2G to evaluate additional derived failure modes originated from Random Hardware failures occurring during boot process, which could affect any Safety functionality within the chip.

5.3 FTA

A Fault Tree Analysis (FTA) in accordance with ISO 26262-5, chapter 7.4.3 has been performed.

Results of this qualitative FTA are:

- Understanding the interaction of architecture elements and safety mechanisms by documentation in fault tree
- Identification of independence requirements
- Indication of potential dependent and common cause failures
- Qualitative ranking of failure directness by minimal cut set analysis

5.4 DFA

A Dependent Failure Analysis (DFA) of the AURIX device(s) was performed for the devices listed in Chapter 3. The different types of dependent failures are assessed inside the DFA according to ISO 26262-9, chapter 7.

All dependent failures are analyzed and safety measures to control or avoid effects of potential failures were identified. FMEDA is used to assess significant failure contributors not covered by a safety mechanism. For details refer to AURIX FMEDAs ([1-4] to [1-12]).

Freedom from interference is proven for AURIX derivatives in accordance with criteria for coexistence as stated in ISO26262-9, clause 6.2. All safety related outcomes of DFA investigations are incorporated into FMEDA or Safety Manual [1-2].

5.5 FMEA

A Failure Mode and Effects Analysis (FMEA) was performed on Design level (DFMEA) including analysis of Startup phase of AURIX 2G to evaluate additional derived failure modes originated from availability and non-availability of Safety Mechanisms during boot phase.

5.6 FMEDA

A quantitative Failure Mode Effect and Diagnostic Analysis (quantitative FMEA) according to ISO 26262-9, chapter 8, has been performed. FMEDA is the selected inductive analysis method. Automated mapping is used to derive the IEC 61508 metrics from ISO 26262 FMEDA.

5.6.1 Base Failure Rate Hard Errors

The failure rate in SN 29500 or IEC TR 62380 [2-2] are considered as Base Failure Rates for Hard Errors. Recognized industry sources require, as input, application specific operation conditions. These are in general unknown to Infineon. Therefore, Infineon assumes application conditions (see Chapter 5.6.1.1.1 Temperature and Mission Profiles (Motor Control)) to calculate the base failure rate. Infineon supports the integrator to extract the hardware part failure rates data from the recognized industry source SN29500 or IEC TR 62380 [2-2]. However, Infineon cannot grant the correct usage nor the correct content of these industry standards.

However, in any case, customer shall validate the provided base failure rate calculation to grant usage consistency of the recognized industry source across all components within the item. This is required as the recognized industry source contain ambiguous rules for usage. In addition, the integrator shall verify the correctness of the calculation and operation condition used within calculation.

The base failure rate is per default calculated according to IEC TR 62380 [2-2]. Please note the following interpretation used in calculation of base failure rates

- Electrical overstress (EOS): For base failure rate calculation according to ISO26262-11:2018, section 4.6.2.1.1, λ_{EOS} is set to 0 FIT as Infineon standard approach because EOS is considered to be a systematic failure mode. Therefore, it is not relevant for FMEDA calculation
- Solder joints: The package failure rate $\lambda_{package}$ calculated according to ISO26262-11:2018 corresponds to the failure modes inside of the package itself (including e.g. the connection between the die and the lead frame) but it also includes the failure rate related to the connection between the package connection points and the board (solder joints). Moreover, the soldering process performed by the system integrator (customer) might deviate from the assumed soldering process being basis for the base failure rate calculation in ISO 26262-11:2018. In such a case, the system integrator shall be responsible for an appropriate consideration of solder joint failure rates. The failure rate of the hardware component calculated in SN 29500-2 includes the failure modes related to the die and to the package however unlike the model described in ISO26262-11:2018, section 4.6.2.1.1 it does not include the failure rate of the connection between the package connection points and the board which is treated separately in SN 29500-5.

Note: IEC TR 62380 is withdrawn and replaced by ISO26262-11 4.6.2 for semiconductors hence reference to IEC TR 62380 and ISO 26262 part 11 are used interchangeably in this document.

5.6.1.1 TC38x Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.1.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 5 Temperature and mission profile TC389 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	

Subject	Information	Comment
Self heating	33.6 K	$P_D[W] \cdot R_{thja}[K/W]$ For $P_D[W]$ and $R_{thja}[K/W]$ values, See [1-15]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.1.2 Base Failure Rate Hard Errors

Table 6 Base Failure Rate TC389 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.59 FIT	
$\lambda_{Package}$	524.76 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	525.35 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.1.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 7 Component parameters TC389 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	395538012 transistors	Extracted by script from net list
Package	PG-LFBGA-516-5	Value from datasheet; package variant used for calculation is PG-LFBGA-516-5 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	

Subject	Information	Comment
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	25.1 mm	Package length and width [2-2]
D (Package)	35.5 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	29.35	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.2 TC39x Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.2.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 8 Temperature and mission profile TC399 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	39.284 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, see [1-16]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.2.2 Base Failure Rate Hard Errors

Table 9 Base Failure Rate TC399 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.98 FIT	
λ_{Package}	535.89 FIT	
$\lambda_{\text{overstress}}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{\text{overstress}}$ can assumed to be 0
Sum λ_{Die} and λ_{Package}	536.88 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{\text{on}} + \tau_{\text{off}} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.2.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 10 Component parameters TC399 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	816321312 transistors	Extracted by script from net list
Package	PG-LFBGA-516-5	Value from datasheet; package variant used for calculation is PG-LFBGA-516-5 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	25.1 mm	Package length and width [2-2]
D (Package)	35.5 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	29.35	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.3 TC35x Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.3.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 11 Temperature and mission profile TC357 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	49.56 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-17]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.3.2 Base Failure Rate Hard Errors

Table 12 Base Failure Rate TC357 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.77 FIT	
$\lambda_{Package}$	291.65 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	292.42 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.3.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 13 Component parameters TC357 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	414496526 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to

Subject	Information	Comment
		variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.4 TC37xED Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.4.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 14 Temperature and mission profile TC377ED based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	40.81 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-19]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.4.2 Base Failure Rate Hard Errors

Table 15 Base Failure Rate TC377ED based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.62 FIT	
λ_{Package}	282.79 FIT	
$\lambda_{\text{overstress}}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{\text{overstress}}$ can assumed to be 0
Sum λ_{Die} and λ_{Package}	283.41 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{\text{on}} + \tau_{\text{off}} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.4.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 16 Component parameters TC377ED based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	404052494 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]

Subject	Information	Comment
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.5 TC37xPD Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.5.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 17 Temperature and mission profile TC377PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	35.2 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-18]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.5.2 Base Failure Rate Hard Errors

Table 18 Base Failure Rate TC377PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.46 FIT	
$\lambda_{Package}$	277.04 FIT	

Subject	Information	Comment
$\lambda_{\text{overstress}}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{\text{overstress}}$ can assumed to be 0
Sum λ_{Die} and λ_{Package}	277.49 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{\text{on}} + \tau_{\text{off}} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.5.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 19 Component parameters TC377PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	216389287 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]

Subject	Information	Comment
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.6 TC36x Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.6.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 20 Temperature and mission profile TC367 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	34.39 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-20]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.6.2 Base Failure Rate Hard Errors

Table 21 Base Failure Rate TC367 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.42 FIT	
$\lambda_{Package}$	276.20 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	276.62 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and $\pi\tau$ are used for calculation of λ_2 term”

5.6.1.6.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 22 Component parameters TC367 based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	150957811 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.7 TC33xED Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.7.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 23 Temperature and mission profile TC337EXT based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	39.56 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-21]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.7.2 Base Failure Rate Hard Errors

Table 24 Base Failure Rate TC337EXT based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.46 FIT	
$\lambda_{Package}$	281.51 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	281.97 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.7.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 25 Component parameters TC337EXT based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	

Subject	Information	Comment
Component technology	MOS: Standard circuits	
Transistor Count	182975433 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.8 TC33xPD/TC32x Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.8.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template. The numbers provided below are valid for TC337 and for TC32x with the same package variant.

Table 26 Temperature and mission profile TC337PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	21.62 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-21]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.8.2 Base Failure Rate Hard Errors

Table 27 Base Failure Rate TC337PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.27 FIT	
$\lambda_{Package}$	262.87 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	263.14 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.8.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 28 Component parameters TC337PD based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	79635275 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to

Subject	Information	Comment
		variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.9 TC3Ex Base Failure Rate Calculation (IEC TR 62380 based)

The base failure rate is per default calculated according to IEC TR 62380 [2-2], see following chapters for details.

5.6.1.9.1 Temperature and Mission Profiles (Motor Control)

The base failure rate calculation is done according to specific use conditions. In ISO26262-10 setting τ_{off} to 0 is suggested to get a conservative calculation of the temperature de-rating factor. However in IEC TR 62380 [2-2] it is explicitly allowed to use powered and unpowered phases of operation to calculate the base failure rate. In the following calculation τ_{off} is not set to 0. Temperature and cycle phase mission profiles provided by IEC TR 62380 [2-2] are used as base for the IFX calculation template.

Table 29 Temperature and mission profile TC3Ex based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Self heating	42.84 K	$P_D[W] \cdot R_{thj-a}[K/W]$ For $P_D[W]$ and $R_{thj-a}[K/W]$ values, See [1-22]
Temperature mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

Subject	Information	Comment
Cycling phases mission profile	Motor control	Mission profiles for automotive; see IEC TR 62380 table 11 [2-2]

5.6.1.9.2 Base Failure Rate Hard Errors

Table 30 Base Failure Rate TC3Ex based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
λ_{Die}	0.595 FIT	
$\lambda_{Package}$	284.855 FIT	
$\lambda_{overstress}$	0 FIT	Due to the application is not a specific interface defined in the IEC TR 62380 chapter 7.3 [2-2], $\lambda_{overstress}$ can assumed to be 0
Sum λ_{Die} and $\lambda_{Package}$	285.45 FIT	

“Note: This calculation is based on the Infineon IEC TR 62380 2004-08: 7.3 calculation default method of λ considering working and dormant times for calculation of τ_i , $\tau_{on} + \tau_{off} = 1$. In case of multiple circuit families assigned to one die, the maximum circuit family values of λ_2 and πt are used for calculation of λ_2 term”

5.6.1.9.3 Component Parameters

Component technology and complexity is heavily influencing the failure rate. The reliability data handbook IEC TR 62380 provides a collection of failure rates or data that is used to calculate failure rates based on field data. Calculating the base failure rate according to IEC TR [2-2], component material and the number of transistors of the IC are taken into account.

Table 31 Component parameters TC3Ex based on IEC TR 62380

Subject	Information	Comment
Calculation Method	IEC TR 62380	
Component technology	MOS: Standard circuits	
Transistor Count	425392804 transistors	Extracted by script from net list
Package	PG-LFBGA-292-6	Value from datasheet; package variant used for calculation is PG-LFBGA-292-6 (because of availability in template), but differences to variant xx-1 are negligible in this case
Substrate material	Epoxy Glass (FR4, G-10)	
Component (Package) material	Epoxy (Plastic package)	
λ_1 Transistor Failure rate	$3.4 \cdot 10^{-6}$	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]

Subject	Information	Comment
λ_2 Technology related failure rate	1.7	Values for MOS standard circuits: Digital circuits, Micros, DSP; see IEC TR 62380 table 16 [2-2]
L and W (Package)	17.1 mm	Package length and width [2-2]
D (Package)	24.18 mm	Package Diagonal as a function of package width and length
λ_3 Base failure rate of the IC package	15.4	Values for surface mounted integrated circuits packages as a function of D (package diagonal); see IEC TR 62380 table 17b [2-2]
α_s (Substrate)	16	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
α_c (Component)	21.5	Thermal expansion coefficients α_s and α_c ; see IEC TR 62380 table 14 [2-2]
π_α thermal expansion factor	1.052	See IEC TR 62380 chapter 7.3.1 [2-2]

5.6.1.10 Base Failure Rate (SN 29500-2 based)

SN29500-2 related base failure rate calculation can be done on request, Infineon can provide support here.

5.6.2 Confidence level – Random Hardware Failures

The Hard error FIT rates in AURIX FMEDAs ([1-4] to [1-12]) are calculated based on recognized industry standard (IEC TR 62380 [2-2] or SN29500). Infineon cannot claim the confidence level for the listed standards.

5.6.3 Base Failure Rate Soft Errors

Soft error rate is composed from NSER and ASER data with a package related adjustment factor.

- ASER is an alpha particle originated soft error rate obtained during accelerated testing and caused by impurities of process and package materials, e.g. mainly due to mold compound and is referenced to low alpha materials. It can also be caused by other materials as solder bumps
- NSER is a soft error rate caused through natural radiation from cosmic rays at earth's surface, obtained during accelerated neutron testing. The value depends on the altitude and the location on earth and is referenced to "New York Sea Level"

$$SER = \left(\frac{\text{Neutron Particle Flux}}{\text{Neutron Particle Flux}_{ref}} \times NSER \right) + \left(\frac{\text{Alpha Particle Flux}}{\text{Alpha Particle Flux}_{ref}} \times ASER \right)$$

$$\text{Alpha Particle Flux}_{ref} = 10 \frac{1}{m^2 h}$$

For low-alpha mold compound,

$$\text{Particle Flux}_{Low-alpha} \leq \text{Particle Flux}_{ref}$$

All AURIX microcontrollers uses low-alpha mold compound (alpha particle emission is 0.0010 cph/cm²) for the packages. Hence the following equation for the SER is applied in the FMEDA.

$$SER = NSER + ASER$$

The Soft error rate is independent from mission profile.

Different values are measured for Flip-Flops (FF) and SRAM are given below.

Flip-Flops:

Alpha SE-rate:	0.605 FIT/Kb
Neutron SE-rate:	0.605 FIT/Kb
Total Flip-Flop SE-rate:	1.211 FIT/Kb

SRAM:

Alpha SE-rate:	0.273 FIT/Kb
Neutron SE-rate:	0.166 FIT/Kb
Total SRAM SE-rate:	0.439 FIT/Kb

5.6.3.1 Confidence level – Soft errors

The soft error rates are measured at reference location and condition as specified in the JESD89 standard i.e. New York City sea level. The confidence level for the Soft Error rate is 90%.

5.6.3.2 Scaling of Soft error rate

Infineon provides option in the FMEDA to scale neutron SER and alpha SER to other altitudes/locations. The customer shall use the FMEDA configuration to scale the $nSER$ value to a particular altitude acc. to JESD89 Annex A and the αSER according to the encapsulation used (scaling is applicable and needed for Bare Die only)

5.6.4 FMEDA results

The FMEDA detailed results are provided in the referenced FMEDAs ([1-4] to [1-12]) for the AURIX Family.

6 Safety Analysis Conclusion

- All safety analysis activities defined in chapter 4 were performed.
- ASIL-B, C, D Hardware Architectural Metrics derived from Top Level Safety Requirements can be fulfilled at application level for the microcontroller under the assumed hardware environment.
- A Dependent Failure Analysis (DFA) for all required independences, covering all functional use cases defined by the Aurix 2G Safety Concept was executed. All findings are addressed in final quantitative safety analysis. Requirements to mitigate Dependent Failure Initiators within application dependent use cases are communicated via ESM requirements through the Safety Manual [1-2].

7 Reference Documents

This section summarizes details of the documents referenced within this report.

7.1 Infineon Documents

Table 32 provides details of the safety relevant documents authored by Infineon and referenced within this report.

Table 32 Infineon Reference Documents

No	Long name / Title	Version	Comment
1-1	Safety Analysis Summary Report	V1.17	This document
1-2	AURIX™ TC3xx Safety Manual	See Safety Package Release Note	Customer document
1-3	AURIX™ TC3xx Safety Manual IEC61508 Addendum	See Safety Package Release Note	Customer document
1-4	AURIX™ TC38x FMEDA	See Safety Package Release Note	Customer document
1-5	AURIX™ TC39x FMEDA	See Safety Package Release Note	Customer document
1-6	AURIX™ TC35x FMEDA	See Safety Package Release Note	Customer document
1-7	AURIX™ TC37xEXT FMEDA	See Safety Package Release Note	Customer document
1-8	AURIX™ TC37xPD FMEDA	See Safety Package Release Note	Customer document
1-9	AURIX™ TC36x FMEDA	See Safety Package Release Note	Customer document
1-10	AURIX™ TC33xEXT FMEDA	See Safety Package Release Note	Customer document
1-11	AURIX™ TC33xPD and TC32x FMEDA	See Safety Package Release Note	Customer document
1-12	AURIX™ TC3Ex FMEDA	See Safety Package Release Note	Customer document
1-13	Recommendations against Common Cause Failures in packages (AP32405)	See Safety Package Release Note	Customer document
1-14	AURIX™ TC3xx User's Manual and User's Manual Appendix	See Safety Package Release Note	Customer document
1-15	AURIX™ TC38x Data Sheet	See Safety Package Release Note	Customer document
1-16	AURIX™ TC39x-B Data Sheet	See Safety Package Release Note	Customer document
1-17	AURIX™ TC35x Data Sheet	See Safety Package Release Note	Customer document
1-18	AURIX™ TC37x Data Sheet	See Safety Package Release Note	Customer document
1-19	AURIX™ TC37xEXT Data Sheet Addendum	See Safety Package Release Note	Customer document
1-20	AURIX™ TC36x Data Sheet	See Safety Package Release Note	Customer document
1-21	AURIX™ TC33xAA Data Sheet	See Safety Package Release Note	Customer document
1-22	AURIX™ TC3Ex Data Sheet	See Safety Package Release Note	Customer document

7.2 References to Standards

This document refers to international standards given in Table 33

Table 33 Standard References

No	Short name	Long name / Title	Version
2-1	ISO 26262	ISO 26262:2018, Road Vehicles — Functional Safety	2 nd Edition, 2018
2-2	IEC TR 62380	Reliability data handbook – Universal model for reliability prediction of electronics components, PCB and equipment	Edition 2004-08
2-3	IEC 61508	IEC 61508:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems	Edition 2010

8 Abbreviations

Table 34 summarizes abbreviations used within this report.

Table 34 Abbreviations

Subject	Explanation
ASER	Soft Error (also called single event or transient faults) Rate obtained during accelerated Alpha particle testing
AURIX 2G	AURIX™ 2 nd Generation (TC3xx)
DFA	Dependent Failure Analysis
DFI	Dependent Failure Initiator
FAE	Field Application Engineer
FF	Flip Flops
FIT	Failure in Time
FMEA	Failure Mode and Effects Analysis
FMEDA	Failure Mode Effect and Diagnostic Analysis
FTA	Fault Tree Analysis
FW	Firmware (see also SSW)
HE	Hard Error (also called permanent faults)
IFX	Infineon technologies
LFM	Latent Fault Metric
NSER	Soft Error Rate obtained during accelerated Neutron particle testing
PMHF	Probabilistic Metric for random Hardware Failures
SASR	Safety Analysis Summary Report
SE	Soft Error (also called as transient faults)
SER	Soft Error (also called single event or transient faults) Rate
SPFM	Single Point Fault Metric
SRAM	Static Random Access Memory
SSW	Startup Software (Firmware)

Revision History

Revision	Date	Description of change
1.0	2018-12	Initial version
1.01	2018-12-13	Deleted TC39x references from cover page and Table 32
1.02	2018-12-19	Added TC39x references to cover page and Table 32 Included chapter 5.6.1.2
1.03	2019-01-03	Fixed typo in table heading “Component parameters TC399x based on IEC TR 62380”
1.04	2019-01-07	Fixed referenced version of Safety Manual
1.05	2019-05-03	TC39x Bare die removed from the scope of this document, as there is no safety case for TC39x bare die yet. Report updated to include TC35x
1.06	2019-07-03	Reference Documents: TC35x FMEDA version number corrected V1.0 → V1.1
1.07	2019-08-13	Report updated to include TC37xED
1.08	2019-09-26	Report updated to include TC37xPD
1.09	2019-12-16	Report updated to include TC36x
1.10	2020-01-27	Report updated to include TC38xA17
1.11	2020-02-17	Report updated to include additional packages for TC37x (ED+PD)
1.12	2020-03-10	Reference to Safety Manual Version has been updated
1.13	2020-03-12	Report updated to include TC33xEXT
1.14	2020-05-29	Report updated to include TC33xPD and TC32x marking option
1.15	2020-06-05	Chapter 5.6.1 updated to include explanation for EOS and solder joint failure rate
1.16	2020-08-18	Report updated to include TC3Ex ISO26262 reference changed to 2 nd Edition, 2018
1.17	2021-05-07	Changed confidentiality to “restricted” Changed reference document version to “See Safety Package Release Note” Added IEC 61508 related safety analyses and reference to AURIX™ TC3xx Safety Manual IEC61508 Addendum Fixed several typos

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